# ENTERPRISE AI PLATFORM RANKING ENGINE

## Extremely Detailed Product Specification

A sector-aware, evidence-weighted AI vendor ranking application that produces a ranked shortlist in under two minutes with granular backend review.

| Field | Value |
|---|---|
| Product | Enterprise AI Platform Ranking Engine |
| Owner | AnalystGenius |
| Version | Product Specification v1.0 |
| Date | 6 May 2026 |
| Status | Draft for product and engineering alignment |
| Source basis | Synthesised from Enterprise AI Assessment Framework v2.0 and subsequent product-model refinement. |
| Core decision | Use the v2.0 12-domain framework as backend methodology; expose a six-pillar ranking UX to users. |

**Product Thesis**
The product should behave less like a questionnaire and more like a ranking engine: minimal user input, maximum backend inference, transparent ranked output, and deep reviewability when required.

# 1. Executive Summary

This product specification defines a production-grade application for ranking enterprise AI platforms. The application is designed to take a small number of high-signal user inputs, apply sector-aware and risk-aware scoring, combine public/proxy/vendor/agent-derived evidence, and return a ranked vendor shortlist with confidence levels, risk flags, evidence drill-down, and recommended validation steps.

The product is deliberately not a full enterprise due-diligence system in its default mode. It uses the full assessment framework in the backend, but presents a simplified executive-ready output to the user. The deeper framework remains available through advanced review, export, vendor-validation workflows, and audit packs.

- Primary user outcome: ranked vendor shortlist in under two minutes.
- Primary product advantage: sector-aware scoring rather than generic vendor comparison.
- Primary methodological advantage: evidence-weighted rankings rather than vendor-claim scoring.
- Primary defensibility layer: backend mapping to security, governance, permissioning, reliability, cost, adoption, vendor resilience, market strength, and capital dependency logic.
- Primary commercial positioning: bridge the gap between static analyst reports and slow bespoke consulting assessments.

**Key Architectural Decision**
v2.0 is the engine room. The six-pillar model is the dashboard. Everything “demoted” from first-screen UX must remain in backend scoring, advanced review, or export mode.

# 2. Target Outcomes and Success Criteria

| Target outcome | Product requirement | Success measure |
|---|---|---|
| Two-minute input | User must complete input flow in <= 2 minutes. | Median completion time <= 120 seconds; abandonment < 20% after first screen. |
| Ranked outcome | App must return ranked vendors with score, confidence, rationale, and validation steps. | At least 3 ranked vendors shown where vendor universe permits. |
| Comprehensive criteria | Backend must preserve full framework coverage, even if simplified in UI. | All six pillars map to detailed backend domains and subfactors. |
| Granular review | Reviewer must be able to inspect how scores were calculated. | Every score links to subfactors, evidence type, source date, and confidence. |
| Evidence integrity | Vendor claims must not score the same as verified evidence. | All scored evidence classified E0-E5 with confidence modifier. |
| Industry differentiation | Ranking must change based on industry archetype and use case. | Industry weight profiles applied and visible in output explanation. |
| Production feasibility | MVP must be buildable with ordinary web app architecture and staged data sources. | No mandatory primary sourcing required for default result. |
| Commercial trust | Output must avoid overclaiming and show uncertainty. | Confidence score, missing evidence, and validation steps shown for every vendor. |

# 3. Product Non-Goals

- The product is not a legal compliance certification tool.
- The product is not a substitute for procurement, security, legal, privacy, or contract review.
- The product is not intended to prove buyer-specific ROI without pilot telemetry.
- The product is not intended to fully validate factuality on private enterprise data in the default flow.
- The product should not make definitive claims that a vendor is objectively best across all contexts.
- The product should not expose the full 12-domain assessment as the default user experience.

# 4. Source Basis and Methodology Backbone

The product uses the Enterprise AI Assessment Framework v2.0 as the methodology backbone. That framework defines 12 assessment domains, score definitions, decision thresholds, practical tests, pass/fail criteria, automatic fail conditions, and regulatory/standards mappings. It also includes industry overlays and capital-resilience thinking. The application should not remove this depth; it should operationalise it behind a lighter UX.

| Methodology asset | Backend use | User-facing use |
|---|---|---|

| | | |
|---|---|---|
| 12-domain framework | Detailed scoring subfactors and advanced review. | Compressed into six pillars. |
| 0-5 scoring model | Capability scoring baseline. | Translated into 0-100 pillar scores. |
| Decision thresholds | Risk/recommendation bands. | Shown as recommendation status: Reject, Pilot, Controlled, Enterprise-scale. |
| Automatic fail conditions | Risk engine blockers and penalties. | Shown as simplified red/amber risk flags. |
| Evidence pack | Evidence schema and vendor validation checklist. | Shown as missing evidence and recommended next steps. |
| Regulatory mappings | Compliance crosswalk and enterprise export pack. | Hidden from default UX; exposed in advanced/export. |
| Industry overlays | Dynamic weights, blockers, and confidence modifiers. | Used in ranking rationale. |
| Capital resilience domain | Vendor-resilience scoring and infrastructure dependency risk. | Shown in Vendor Resilience pillar. |

## 5. Primary Personas

| Persona | Primary goal | Needs | Product value |
|---|---|---|---|
| CEO / COO / Board sponsor | Decide which AI platforms deserve serious consideration. | Fast answer, confidence, risk summary, defensible rationale. | Ranked shortlist and executive scorecard. |
| CIO / CTO / CDO | Assess fit with enterprise architecture and data strategy. | Integration, security, scalability, model portability, observability. | Drill-down into Enterprise Control and Integration & Operations. |
| CISO / Security architect | Identify unsafe platforms before procurement advances. | Permissioning, data flow, subprocessors, threat resilience, auditability. | Risk flags, evidence sources, advanced security detail. |
| Procurement / Vendor management | Compare vendors quickly and identify validation asks. | Evidence pack, risk gaps, commercial resilience, exit risk. | Vendor comparison and missing-evidence checklist. |
| AI governance / Risk / Legal | Ensure deployment aligns with governance and regulatory expectations. | Use-case risk, auditability, human oversight, standards mapping. | Advanced compliance export and risk register. |
| Business unit lead | Find practical platforms for specific workflows. | Use-case fit, adoption likelihood, business value, ecosystem fit. | Business Fit score and recommended validation tasks. |

## 6. End-to-End User Journey

1. Landing page explains the product promise: answer 10 questions, receive ranked AI platform shortlist with evidence-backed scoring.
2. User selects industry archetype and optionally region/jurisdiction.
3. User selects organisation size and enterprise maturity stage.
4. User selects primary AI objective and one or more use cases.
5. User sets risk profile, data sensitivity, autonomy appetite, deployment preference, budget sensitivity, and current ecosystem.
6. User chooses vendors to compare or asks the system to recommend a shortlist.
7. App retrieves vendor intelligence profiles and applies dynamic weighting.
8. Scoring engine calculates pillar scores, confidence scores, risk penalties, missing-evidence penalties, and sector adoption fit.
9. User receives ranked shortlist with explanation, risk flags, confidence, missing evidence, and next validation steps.
10. User can open advanced drill-down for each score, export a board pack, or start a deeper vendor-validation workflow.

## 7. Two-Minute Input Flow Specification

| Step | Input | Control type | Required? | Backend effect |
|---|---|---|---|---|
| 1 | Industry archetype | Dropdown/search | Yes | Applies industry weights, blockers, adoption maturity, evidence strictness. |
| 2 | Region/jurisdiction | Dropdown/multi-select | Optional MVP; important v2 | Adds EU/UK/US/regional privacy and regulatory modifiers. |
| 3 | Organisation size | Dropdown | Yes | Adjusts scale, control, integration, and vendor maturity expectations. |
| 4 | AI maturity stage | Dropdown | Optional MVP | Adjusts urgency and validation recommendations. |
| 5 | Primary objective | Multi-select | Yes | Adjusts Business Fit and Market Strength interpretation. |

| | | | | |
|---|---|---|---|---|
| 6 | Main use case(s) | Multi-select | Yes | Maps to risk tier, vendor capability fit, reliability needs. |
| 7 | Data sensitivity | Slider 1-5 | Yes | Adjusts Enterprise Control weight and blocker severity. |
| 8 | Risk tolerance | Slider 1-5 | Yes | Adjusts penalty thresholds and recommendations. |
| 9 | Autonomy appetite | Dropdown | Yes | Adjusts Agentic/autonomy controls and risk penalties. |
| 10 | Existing ecosystem | Multi-select | Yes | Adds integration fit bonus/penalty. |
| 11 | Deployment preference | Dropdown | Yes | Adjusts SaaS/VPC/on-prem/sovereign fit. |
| 12 | Budget sensitivity | Slider 1-5 | Yes | Adjusts Cost/TCO weighting and tolerance. |
| 13 | Vendor selection | Select vendors or recommend | Yes | Defines comparison set. |

**MVP Input Rule**
The MVP should not ask users to manually score vendors. The user supplies context; the backend supplies vendor scoring.

# 8. Six-Pillar Product Model

## 8.x Business Fit

Default weight: 15%

Determines how well the vendor matches the user's industry, use case, objective, workflow, organisation size, maturity stage, and adoption context.

| Backend mapping | Evidence/proxy signals |
|---|---|
| Strategic value, Use-case fit, Workflow relevance, Industry fit, Ecosystem relevance, Workforce adoption likelihood, Industry AI adoption maturity | Use-case coverage, Sector-specific proof, Pilot-to-production relevance, Workflow depth, User adoption proxy |

## 8.x Enterprise Control

Default weight: 25%

Determines whether the platform can be safely governed inside an enterprise environment.

| Backend mapping | Evidence/proxy signals |
|---|---|
| Data security, Privacy, Retention, Subprocessors, Identity, Permissioning, Governance, Auditability, Compliance | Training-data policy, SSO/SCIM, Source-permission inheritance, Audit logs, Data residency, Policy enforcement |

## 8.x Reliability & Safety

Default weight: 15%

Determines whether the platform produces grounded, accurate, safe, repeatable, and appropriately uncertain outputs.

| Backend mapping | Evidence/proxy signals |
|---|---|
| Factuality, Hallucination control, Citation quality, Conflict handling, Threat resilience, Red-team response | Baseline factuality, Citation validity, Refusal of unsupported answers, Prompt-injection resistance, Adversarial behaviour |

## 8.x Integration & Operations

Default weight: 15%

Determines whether the platform can fit into the buyer's existing systems and operate predictably at scale.

| Backend mapping | Evidence/proxy signals |
|---|---|
| Connectors, API quality, Architecture fit, Model portability, Agentic controls, Cost at scale, FinOps, Observability | Connector breadth, API maturity, Cost risk, 3x usage sensitivity, Latency/availability signals, Sandbox support |

### 8.x Vendor Resilience

Default weight: 15%

Determines whether the vendor can survive, scale, support, and honour the relationship over time.

| Backend mapping | Evidence/proxy signals |
|---|---|
| Vendor maturity, Lock-in, Exitability, Capital resilience, Infrastructure dependency, Investor/IPO/acquisition risk | Funding/runway proxy, Model/cloud/GPU dependency, Export support, SLA/support maturity, Strategic partner concentration |

### 8.x Market Strength

Default weight: 15%

Determines whether the vendor is competitively relevant, adopted, differentiated, and strategically credible.

| Backend mapping | Evidence/proxy signals |
|---|---|
| Market share, Vendor adoption by industry, Market momentum, Deployment depth, Strategy, Differentiation, Ecosystem traction | Category share, Sector customer evidence, Review/community/job signals, Roadmap credibility, Partner ecosystem |

## 9. Backend Domain Mapping

| Backend domain | Primary pillar | Secondary pillar | Role in scoring |
|---|---|---|---|
| Strategic Value & Use-Case Fit | Business Fit | Market Strength | Determines whether vendor capabilities align to the selected AI objective and use case. |
| Data Security, Privacy & Retention | Enterprise Control | Vendor Resilience | Scores training use, retention, deletion, residency, subprocessors, DLP, and policy controls. |
| Identity, Permissioning & Access Control | Enterprise Control | Integration & Operations | Scores SSO, SCIM, RBAC/ABAC, source-permission inheritance, and permission-laundering risk. |
| Model Reliability, Factuality & Hallucination | Reliability & Safety | Business Fit | Scores baseline factuality, citation behaviour, uncertainty, conflict handling, numerical reliability. |
| Governance, Compliance & Auditability | Enterprise Control | Reliability & Safety | Scores inventory, risk-tiering, decision reconstructability, human oversight, regulatory support. |
| Security Architecture & Threat Resilience | Reliability & Safety | Enterprise Control | Scores prompt injection defence, data exfiltration controls, API security, pen-test evidence, incident response. |
| Integration & Architecture Fit | Integration & Operations | Business Fit | Scores connectors, APIs, sandboxing, model portability, graceful degradation, async support. |
| Agentic Capability & Autonomy Controls | Integration & Operations | Reliability & Safety | Scores approval gates, tool scoping, action logs, reversibility, simulation mode, kill switch. |
| Cost, TCO & FinOps | Integration & Operations | Vendor Resilience | Scores pricing clarity, usage volatility, unit economics, caps, chargeback, 3x usage risk. |
| Workforce Adoption & Change Management | Business Fit | Market Strength | Scores training, acceptable-use policy, adoption support, user friction, overtrust/undertrust risk. |
| Vendor Maturity, Lock-In & Exit | Vendor Resilience | Integration & Operations | Scores platform depth, model portability, exportability, migration support, deprecation policy. |
| Capital Resilience & Strategic Dependency | Vendor Resilience | Market Strength | Scores runway proxy, investor conflict, infrastructure dependency, IPO/acquisition disruption, customer concentration. |
| Market Position, Adoption & Vendor Strategy | Market Strength | Business Fit | Scores category share, sector adoption, deployment depth, market momentum, roadmap credibility, differentiation. |

## 10. Industry Differentiation Specification

Industry is a primary context variable. It changes weightings, blockers, evidence standards, confidence penalties, and output explanations. The same vendor may rank differently for a bank, hospital, law firm, SaaS company, public sector body, or manufacturer.

| Industry archetype | Default weight profile | Fatal or severe blockers |
|---|---|---|

| | | |
|---|---|---|
| Regulated financial enterprise | Business 10 / Control 35 / Reliability 20 / Ops 10 / Resilience 15 / Market 10 | Weak auditability; weak permissioning; poor data residency; no model/version tracking; weak operational resilience; no exit plan. |
| Health & life sciences | Business 10 / Control 30 / Reliability 25 / Ops 10 / Resilience 15 / Market 10 | Unsupported clinical claims; weak PHI/privacy controls; no human oversight for patient-impacting workflows; weak traceability. |
| Legal & professional advisory | Business 15 / Control 30 / Reliability 20 / Ops 10 / Resilience 10 / Market 15 | Hallucinated citations; poor privilege controls; no matter/client permissioning; unclear training policy. |
| Public sector & education | Business 10 / Control 35 / Reliability 15 / Ops 10 / Resilience 15 / Market 15 | Weak transparency; poor records retention; unclear residency; weak accessibility; no public accountability trail. |
| Critical infrastructure & defence | Business 5 / Control 40 / Reliability 20 / Ops 10 / Resilience 20 / Market 5 | Single cloud/model dependency; foreign jurisdiction conflict; no sovereign/private option; weak supply-chain transparency. |
| Enterprise software & digital product | Business 20 / Control 15 / Reliability 15 / Ops 20 / Resilience 10 / Market 20 | Poor APIs/SDKs; no model portability; weak uptime/rate limits; weak customer-facing AI safety. |
| Industrial & physical operations | Business 15 / Control 25 / Reliability 20 / Ops 20 / Resilience 15 / Market 5 | Unreliable operational-system integration; weak offline/edge support; poor safety validation. |
| Commercial enterprise | Business 20 / Control 20 / Reliability 15 / Ops 15 / Resilience 10 / Market 20 | Weak customer-data controls; poor brand safety; weak cost predictability; poor adoption UX. |

# 11. Industry AI Adoption Percentage and Vendor Adoption Logic

The product must distinguish industry-level AI maturity from vendor-specific adoption in that industry. High general AI adoption does not prove a vendor is good. Vendor sector adoption does not override weak security or reliability. Adoption is a context and confidence modifier, not a quality substitute.

| Metric | Definition | Use in scoring |
|---|---|---|
| Industry experimentation adoption % | Share of organisations trialling or experimenting with AI. | Weak maturity signal; informs market curiosity. |
| Industry regular-use adoption % | Share using AI in at least one business function. | Moderate maturity signal. |
| Industry production adoption % | Share with live production AI use cases. | Strong maturity signal; feeds industry maturity score. |
| Industry scaled adoption % | Share scaling AI across multiple functions. | Strongest maturity signal; highest weight. |
| Agentic experimentation % | Share trialling AI agents. | Adjusts autonomy market readiness. |
| Agentic scaled adoption % | Share scaling AI agents. | Adjusts autonomy confidence and risk expectations. |
| Vendor sector deployment depth | Evidence of vendor production use in selected industry. | Feeds Market Strength and Business Fit. |

| Adoption maturity band | Percentage range | Model interpretation |
|---|---|---|
| Nascent | 0-20% | High change-management friction; early-mover upside possible. |
| Emerging | 21-40% | Vendor proof and adoption support matter heavily. |
| Developing | 41-60% | Adoption growing; fit and governance matter. |
| Mainstream | 61-80% | Differentiation and deployment quality matter. |
| Advanced | 81-100% | Focus on quality, control, scale, and vendor maturity. |

**Adoption Calculation**

Industry Maturity Modifier = 25% regular-use adoption + 35% production adoption + 40% scaled adoption. Adoption Friction Penalty = 100 - Industry Adoption Maturity Score, applied modestly and adjusted by user objective.

# 12. Scoring Engine Specification

## 12.1 Core Score Objects

| Object | Description | Range |
|---|---|---|
| raw_subfactor_score | Capability score before evidence and context adjustment. | 0-100 |
| evidence_modifier | Multiplier based on evidence grade E0-E5. | 0.0-1.0 |
| context_weight | Dynamic weight based on industry, use case, risk, data sensitivity, autonomy, budget, | 0-1 |

| | | |
|---|---|---|
| | ecosystem. | |
| risk_penalty | Deduction applied for red flags or blockers. | 0-100 impact |
| missing_evidence_penalty | Deduction for absent evidence in important areas. | 0-100 impact |
| strategic_fit_bonus | Positive adjustment for strong ecosystem, use-case, sector, or deployment fit. | 0-10 recommended |
| sector_adoption_fit_bonus | Positive adjustment for strong vendor proof in selected industry. | 0-10 recommended |
| confidence_score | Overall evidence strength and freshness score. | 0-100 |

### 12.2 Formula

**Final Vendor Score Formula**

Final Vendor Score = Sum(Pillar Score x Dynamic Context Weight x Evidence Confidence) + Strategic Fit Bonus + Sector Adoption Fit Bonus - Risk Penalties - Missing Evidence Penalty - Adoption Friction Penalty

### 12.3 Evidence Modifiers

| Evidence grade | Meaning | Modifier | Example |
|---|---|---|---|
| E0 | No evidence. | 0.0 | No documentation, no claim, no test. |
| E1 | Vendor claim only. | 0.4 | Marketing page states capability. |
| E2 | Public documentation. | 0.6 | Docs/trust page confirms feature. |
| E3 | Public test, agent test, sandbox/API verification. | 0.75 | Our agent or public API verifies behaviour. |
| E4 | Production customer evidence. | 0.9 | Case study/reference with production deployment. |
| E5 | Independent audit, verified benchmark, third-party validation. | 1.0 | SOC/audit/independent benchmark proves capability. |

## 13. Risk Penalty and Blocker Engine

| Risk class | Definition | Default treatment | Example |
|---|---|---|---|
| Fatal blocker | Risk incompatible with selected context. | Exclude or mark Not Recommended. | Unclear training-data policy for regulated sensitive data. |
| Severe risk | High-risk gap that may not be fatal in all contexts. | Large score penalty and red flag. | Weak permissioning evidence in professional services. |
| Moderate risk | Important but remediable gap. | Moderate penalty and validation step. | Unclear cost at 3x usage. |
| Low risk | Minor uncertainty or missing public proof. | Confidence reduction. | Limited public customer references. |

| Backend fail condition group | Examples | User-facing translation |
|---|---|---|
| Security/privacy/governance | Data-training ambiguity; unauthorised retrieval; no audit trail. | Critical control risk. |
| Reliability/autonomy | Unsupported confident answers; high-impact autonomous actions; no kill switch. | Reliability or autonomy blocker. |
| Commercial/financial | No ROI mechanism; no cost telemetry; no runway evidence. | Commercial resilience risk. |
| Strategic dependency | Single model/cloud/GPU dependency; no export/deletion. | Vendor dependency risk. |
| Market/adoption/strategy | False market leadership; weak adoption evidence; poor roadmap delivery. | Market confidence risk. |

## 14. Data Architecture and Evidence Collection

### 14.1 Data Layers

| Layer | Purpose | Examples |
|---|---|---|
| Static vendor database | Curated baseline profile for each vendor. | Category, supported use cases, integrations, |

| | | |
|---|---|---|
| | | security docs, pricing, known risks. |
| Public data refresh | Keep vendor profiles current. | Docs, trust centres, pricing pages, subprocessors, changelogs, status pages, filings, GitHub, reviews, job postings. |
| LLM-agent scoring | Extract, classify, compare, and score public evidence. | Security maturity, governance maturity, cost risk, adoption proxy, strategy credibility. |
| Optional validation inputs | Upgrade confidence for deeper workflows. | RFI, contracts, SOC reports, telemetry, customer references, vendor questionnaires. |

### 14.2 Data Source Registry

| Source category | Data captured | Typical confidence | Freshness requirement |
|---|---|---|---|
| Vendor docs | Features, APIs, integrations, deployment options. | Medium | 30-90 days. |
| Trust centre | Security certifications, subprocessors, DPA, privacy, retention. | Medium-high | 30-90 days. |
| Pricing pages | Seat/API/token/public packages. | Medium | 30-60 days. |
| Status pages | Availability, incidents, service degradation. | Medium-high | Daily/weekly. |
| Changelogs/release notes | Roadmap delivery and product velocity. | Medium-high | Weekly/monthly. |
| Public filings | Financials, risk disclosures, ownership for public companies. | High for public firms | Quarterly/annual. |
| Job postings | Strategic signals, hiring trends, technical direction. | Low-medium | Weekly/monthly. |
| Review platforms | Customer sentiment and adoption clues. | Low-medium | Weekly/monthly. |
| Marketplaces/app stores | Installs, rankings, ecosystem activity. | Low-medium | Weekly/monthly. |
| GitHub/open source | Developer adoption and release activity. | Medium for dev tools | Weekly. |
| Analyst/public reports | Market share, adoption benchmarks, category trends. | Medium-high if licensed/current | Quarterly/annual. |

## 15. LLM-Agent Backend Specification

LLM agents should not make final decisions. They should perform structured extraction, classification, comparison, gap detection, and evidence summarisation under strict schemas. Deterministic code should calculate final scores from structured agent outputs.

| Agent | Inputs | Outputs | Guardrails |
|---|---|---|---|
| Vendor Evidence Extractor | Vendor pages, docs, pricing, trust centre. | Structured evidence items tagged by pillar/domain/subfactor. | Must include source URL/date and direct excerpt; no unsupported claims. |
| Evidence Classifier | Evidence items. | Evidence grade E0-E5, confidence, freshness, relevance. | Conservative grading; vendor marketing defaults to E1/E2. |
| Security/Governance Agent | Trust docs, DPAs, admin docs. | Security/control maturity sub-scores and gaps. | No “pass” without documented proof. |
| Market/Strategy Agent | News, releases, jobs, partnerships, changelogs. | Strategy credibility, momentum, differentiation. | Distinguish stated strategy from delivered product. |
| Adoption Proxy Agent | Reviews, traffic proxies, job postings, case studies, marketplace signals. | Adoption proxy and deployment-depth indicators. | Label as proxy; never treat as true WAU/MAU. |
| Cost Modelling Agent | Pricing pages, API rates, usage archetypes. | Relative cost-risk score and sensitivity model. | Output relative 1-10 risk, not exact cost unless validated. |
| Risk Flag Agent | All structured scores/evidence. | Potential blockers, penalties, missing evidence. | Human-reviewable; fatal blockers require rule confirmation. |
| Narrative Agent | Final scores and evidence. | User-facing explanation. | Cannot introduce facts not in score object/evidence store. |

## 16. Data Model / Database Schema

| Table / Collection | Purpose | Key fields |
|---|---|---|
| vendors | Canonical vendor entities. | vendor_id, name, category, website, HQ, ownership_type, public_private, summary. |
| vendor_products | Product-level records where vendors have multiple AI offerings. | product_id, vendor_id, product_name, category, supported_use_cases, deployment_models. |
| vendor_evidence_items | Atomic evidence records. | evidence_id, vendor_id, product_id, source_url, captured_at, excerpt, domain, subfactor, evidence_grade, confidence. |
| vendor_scores | Computed scores per vendor/context. | score_id, vendor_id, context_hash, pillar_scores_json, final_score, confidence_score, generated_at. |
| industries | Industry archetypes and weight profiles. | industry_id, name, archetype, default_weights_json, blockers_json. |

| | | |
|---|---|---|
| industry_adoption_profiles | Industry-level adoption maturity data. | industry_id, experimentation_pct, regular_use_pct, production_pct, scaled_pct, agentic_pct, source, source_date, confidence. |
| vendor_industry_adoption | Vendor proof in each industry. | vendor_id, industry_id, customer_count_estimate, production_reference_count, deployment_depth_score, confidence. |
| use_cases | Use-case taxonomy. | use_case_id, name, risk_tier, autonomy_default, reliability_requirement, core_workflows. |
| scoring_rules | Versioned scoring and weighting rules. | rule_id, version, trigger_condition, weight_changes, penalties, blockers. |
| risk_flags | Generated or curated risk flags. | risk_id, vendor_id, context_hash, severity, description, evidence_ids, treatment. |
| runs | User assessment run records. | run_id, user_id, inputs_json, vendor_set, created_at, output_json. |
| audit_log | Internal traceability. | event_id, actor, action, object_type, object_id, timestamp, before_after. |

# 17. API and Service Specification

| Endpoint/service | Method | Purpose | Inputs | Output |
|---|---|---|---|---|
| /api/assessment/start | POST | Create assessment run. | User input context. | run_id, normalised_context. |
| /api/vendors/recommend | POST | Recommend vendor set if user has not selected vendors. | Industry, use_case, ecosystem, deployment preference. | vendor_ids with rationale. |
| /api/assessment/score | POST | Run scoring engine. | run_id, vendor_ids. | ranked_results, pillar_scores, confidence. |
| /api/assessment/{id} | GET | Retrieve assessment result. | run_id. | Full result object. |
| /api/assessment/{id}/explain | GET | Retrieve drill-down explanation. | run_id, vendor_id optional. | Evidence-linked explanation tree. |
| /api/assessment/{id}/export | POST | Generate board/advanced export. | run_id, export_type. | DOCX/PDF/JSON package. |
| /api/vendor/{id}/evidence | GET | Retrieve evidence items. | vendor_id, filters. | Evidence records. |
| /api/admin/vendor/refresh | POST | Trigger public-data refresh. | vendor_id or category. | job_id. |
| /api/admin/evidence/review | POST | Human review evidence grading. | evidence_id, reviewer_decision. | Updated evidence item. |

# 18. Output Schema

| Field | Description |
|---|---|
| run_id | Unique assessment identifier. |
| input_summary | Normalised user inputs and derived context. |
| ranking | Ordered list of vendor result objects. |
| vendor_result.final_score | 0-100 final adjusted score. |
| vendor_result.confidence_score | 0-100 evidence confidence score. |
| vendor_result.recommendation_band | Reject, Pilot Only, Controlled Deployment, Enterprise-Scale Candidate. |
| vendor_result.pillar_scores | Business Fit, Enterprise Control, Reliability & Safety, Integration & Operations, Vendor Resilience, Market Strength. |
| vendor_result.top_strengths | 3-5 user-facing strengths. |
| vendor_result.top_risks | 3-5 user-facing risks. |
| vendor_result.missing_evidence | Evidence gaps ranked by importance. |
| vendor_result.validation_steps | Recommended next validation steps. |
| vendor_result.industry_rationale | Explanation of how industry changed ranking. |
| vendor_result.evidence_links | Evidence IDs used in scoring. |
| comparison_summary | Cross-vendor insight: why first outranks second. |

# 19. UX / Screen Specification

| Screen | Purpose | Required elements |
|---|---|---|
| Landing / value proposition | Explain benefit and start flow. | One-line promise, example output, CTA, trust note. |
| Input step 1: context | Capture industry, size, region, maturity. | Dropdowns, progress indicator, help text. |
| Input step 2: use case | Capture objective/use case/risk tier. | Use-case multi-select, objective selection, examples. |
| Input step 3: constraints | Capture sensitivity, risk, autonomy, ecosystem, budget. | Sliders/dropdowns, tooltips. |
| Vendor selection | Choose vendor set. | Selected vendors, recommended shortlist option, all vendors option. |
| Loading / analysis state | Show backend process without pretending certainty. | Progress stages: context weighting, evidence retrieval, scoring, ranking. |
| Results overview | Show ranked list. | Rank, score, confidence, recommendation band, top |

| | | |
|---|---|---|
| | | reason. |
| Vendor detail | Explain vendor score. | Six pillar chart/table, strengths, risks, missing evidence, validation steps. |
| Advanced drill-down | Expose backend granularity. | Subfactors, evidence, source dates, confidence, penalties. |
| Comparison view | Side-by-side vendor analysis. | Pillar comparison, risk comparison, evidence confidence. |
| Export view | Generate board pack or detailed report. | Export options: summary, advanced, compliance pack, JSON. |

## 20. Backend-Retained Elements Previously Demoted from UX

The following elements must not be removed. They should be hidden from the default user experience but fully retained in backend scoring, advanced review, or export mode.

| Element | User-facing treatment | Backend treatment | Why retained |
|---|---|---|---|
| Full regulatory mapping | Hidden from default result. | Included in compliance crosswalk and export mode. | Needed for enterprise credibility, legal review, standards alignment. |
| 20 automatic fail conditions | Simplified risk flags. | Fully included in risk engine as blockers/penalties. | Prevents unsafe rankings. |
| Full evidence pack | Summarised as missing evidence. | Evidence schema and vendor validation checklist. | Supports confidence and validation workflows. |
| Long analogies | Removed from app scoring UX. | Optional narrative report-writing layer only. | Useful for explanatory reports, not score calculation. |
| 100-300 question gold-set testing | Not in MVP input flow. | Validation/premium test module. | Needed for deployment validation. |
| Full due-diligence questionnaire | Hidden from default flow. | Advanced/vendor validation mode. | Useful for procurement workflows. |
| Clause-level EU/ISO/NIST mapping | Not first output. | Included in backend compliance export. | Needed for audit/legal/board pack. |

## 21. Non-Functional Requirements

| Category | Requirement |
|---|---|
| Performance | Default ranking result should return in <= 20 seconds from cached profiles. Cold evidence refresh may run asynchronously. |
| Scalability | Support vendor universe of at least 500 vendors and 10,000+ evidence items in MVP architecture. |
| Explainability | Every final score must be traceable to pillar, subfactor, evidence, confidence modifier, and penalty/bonus. |
| Reliability | Assessment runs must be deterministic for a given scoring-rule version and evidence snapshot. |
| Versioning | Scoring rules, evidence snapshots, and industry weights must be versioned. |
| Security | Admin tools must be access-controlled. Uploaded validation documents must be isolated and encrypted. |
| Privacy | User inputs should not be used to train models. Uploaded documents must have retention controls. |
| Auditability | All admin score changes and evidence-grade overrides must be logged. |
| Data freshness | Evidence items must show captured_at/source_date and freshness status. |
| Fallback | If data is missing, app should lower confidence and show missing evidence, not fabricate. |

## 22. Testing and Acceptance Criteria

| Test area | Acceptance criteria |
|---|---|
| Input flow | Median test-user completion <= 2 minutes; no input requires expert procurement knowledge. |
| Scoring determinism | Same inputs + same evidence snapshot + same scoring version produce same result. |
| Evidence traceability | Every non-zero subfactor score has evidence or an explicit inferred/proxy marker. |
| No hallucinated evidence | Narrative output cannot mention claims not present in result object/evidence store. |
| Industry differentiation | Same vendor set produces materially different weightings/rationales across at least 4 archetypes. |
| Risk penalties | Known fatal blockers trigger Not Recommended in regulated/high-risk |

| | |
|---|---|
| | contexts. |
| Confidence scoring | Vendor with claims-only evidence scores lower confidence than documented/tested vendor. |
| Drill-down | Reviewer can inspect pillar > subfactor > evidence > source date > confidence. |
| Export | Board summary and advanced export generate without layout breakage. |
| Performance | Cached result generated <= 20 seconds for 10 vendors. |

# 23. MVP Scope

| MVP component | Include in MVP? | Notes |
|---|---|---|
| Two-minute input flow | Yes | Core product requirement. |
| Six-pillar results | Yes | Main user-facing output. |
| 12/13-domain backend mapping | Yes | Needed for granularity. |
| Evidence grading E0-E5 | Yes | Core credibility feature. |
| Dynamic industry weights | Yes | Use 8 archetypes. |
| Industry adoption maturity | Yes | Use manually curated initial dataset; refresh later. |
| Vendor profile database | Yes | Start with curated vendors. |
| Public data scraper | Partial | Begin with scheduled page fetchers and manual review. |
| LLM evidence classifier | Yes | Constrained schema only. |
| Advanced drill-down | Yes | Essential for trust. |
| Board export | Yes | Simple DOCX/PDF export. |
| Full compliance pack | No, v2 | Backend mapping ready; export later. |
| Automated live agent tests | No, v2 | Start with documented/sandbox scoring. |
| Customer-reference workflow | No, v2 | Premium/deep mode. |

# 24. Implementation Sequence

| Phase | Build focus | Deliverables |
|---|---|---|
| Phase 1: Data model and scoring skeleton | Define schemas, six pillars, domain mapping, industry weights, evidence grades. | DB schema, scoring-rule config, static sample vendors. |
| Phase 2: Input and ranking MVP | Build user input flow and deterministic scoring engine. | Working flow returns ranked vendors from seeded data. |
| Phase 3: Evidence and drill-down | Attach evidence items to subfactor scores and expose drill-down. | Evidence traceability, confidence score, missing evidence. |
| Phase 4: Industry/adoption layer | Add industry archetypes, adoption maturity, vendor sector proof. | Industry-adjusted rankings and explanations. |
| Phase 5: Public data ingestion | Build fetchers/scrapers for vendor docs, pricing, trust centres, status pages. | Evidence ingestion queue, source freshness. |
| Phase 6: LLM evidence classifier | Extract and classify evidence into strict schema. | Evidence grading assistant with admin review. |
| Phase 7: Risk engine | Add blockers, penalties, missing-evidence penalties. | Risk flags and Not Recommended logic. |
| Phase 8: Exports and admin | Board pack, advanced report, admin evidence review. | Exportable outputs and governance tooling. |

# 25. Product Backlog

| Epic | Key stories |
|---|---|
| User input | As a user, I can complete context inputs in under two minutes. As a user, I can choose recommended vendors or select my own. |
| Ranking output | As a user, I receive ranked vendors with final score, confidence, and rationale. |
| Pillar scoring | As a user, I can view six pillar scores. As a reviewer, I can expand subfactor detail. |
| Evidence engine | As an admin, I can view, approve, downgrade, or remove evidence items. |
| Industry rules | As a user, I see rankings adjusted by my sector. As admin, I can update industry weights. |
| Adoption logic | As a user, I see whether sector maturity and vendor sector proof |

| | |
|---|---|
| | affected the score. |
| Risk flags | As a user, I see the most important risks. As admin, I can inspect triggered rules. |
| Vendor profiles | As an admin, I can edit vendor capabilities, evidence, and data freshness. |
| Exports | As a user, I can download a board summary or advanced assessment. |
| Audit | As an admin, all score/evidence changes are logged. |

# 26. Open Decisions

- Initial vendor universe: which vendors are included in MVP and which categories are in/out?
- Evidence source policy: which public sources are accepted for production scoring?
- Data licensing: which analyst reports or third-party sources can legally be used?
- Region strategy: should MVP include UK/EU/US jurisdiction modifiers or only industry modifiers?
- Scoring calibration: should weights be manually set at launch or calibrated against sample vendor assessments?
- Export format: should MVP export DOCX, PDF, or both?
- User accounts: anonymous single-run assessments versus authenticated saved projects?
- Vendor self-submission: should vendors be able to submit evidence later?
- Human review: which evidence grades require analyst approval before affecting production scores?
- Pricing model: free teaser, paid report, enterprise subscription, or analyst dashboard?

# 27. Final Recommended Product Shape

**Final Recommendation**
Build a sector-aware, evidence-weighted enterprise AI vendor ranking app. Use a two-minute input flow and a six-pillar output. Keep the full v2.0 framework, fail conditions, evidence pack, regulatory mapping, and industry overlays in the backend and advanced/export layers.

- Do not expose the full 12-domain framework by default.
- Do not remove the full framework from backend logic.
- Do not let vendor claims score as verified evidence.
- Do not hard-code generic rankings detached from user context.
- Do not treat adoption percentage as platform quality.
- Do not overstate final certainty; always show confidence and validation gaps.

# Appendix A. Source Framework Dependencies

- Enterprise AI Assessment Framework v2.0: 12-domain methodology, decision thresholds, score definitions, fail conditions, evidence packs, standards mapping, industry overlays, and capital resilience domain.
- Product-model refinement: six-pillar app UX, two-minute input flow, evidence-weighted scoring, public/proxy data architecture, industry adoption maturity logic, market strength and vendor strategy logic.
- Backend retention decision: full regulatory mapping, automatic fail conditions, full evidence pack, gold-set testing methodology, due-diligence questionnaire, and clause-level mappings remain in backend/advanced/export mode.